

6 Some people who are infected with HIV have HIV/AIDS.

(a) State the full term for HIV.

..... [1]

(b) Following transmission of HIV, early diagnosis of infection and treatment with anti-retroviral therapy (ART) helps to control the spread of the pathogen and prevent HIV/AIDS.

Suggest why treating people who have developed HIV/AIDS with ART may help to reduce the number of overall deaths from infectious diseases, such as cholera, TB and malaria.

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..... [2]

(c) Studies suggest that people who are infected with HIV may be at a higher risk of heart disease. One cause of heart disease is the narrowing of the lumen of one or both of the main coronary arteries.

A coronary artery bypass graft (bypass graft) is a surgical operation that uses healthy blood vessels to divert blood around diseased sections of coronary arteries. The main choice of blood vessel to use for a bypass graft is known as the internal thoracic artery.

Fig. 6.1 is a diagram of an external view of the heart to show a **double** bypass graft.

The blood vessels used in the bypass graft shown in Fig. 6.1 are the great saphenous vein from the leg and the internal thoracic artery.

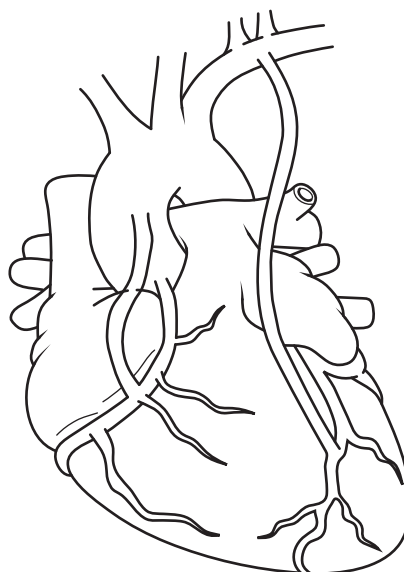


Fig. 6.1



(i) Draw a cross (X) on Fig. 6.1 to show an area of the **right** coronary artery that has been bypassed. [1]

(ii) After surgery, the wall of the great saphenous vein becomes thicker.

Suggest **and** explain why it is important for the wall of the vein to become thicker after surgery.

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..... [2]

[Total: 6]